IGCSE 0607 Practice Questions — E4.2

Gradient of linear graphs

Total Questions: 9 — **Total Marks:** 50 Marks

Table of Contents & Quick Navigation

- **E4.2** Gradient of linear graphs
  → **Paper 2 (Non-Calculator)** *(6 Qs)* · **Paper 4 (Calculator)** *(3 Qs)*

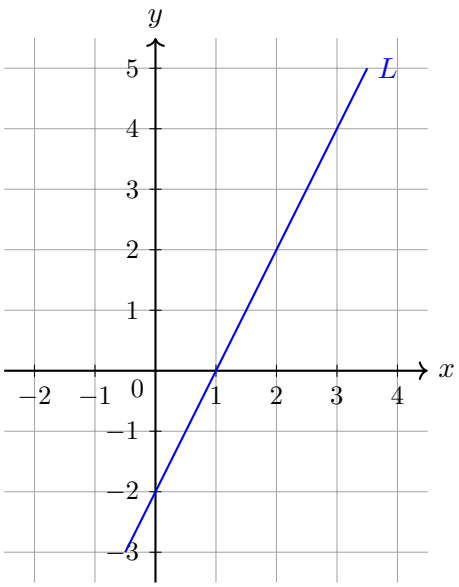
---

1 Points A and B have coordinates $(2, -3)$ and $(6, 5)$ respectively.

Work out the gradient of the line AB.

..... **[2]**

2 The diagram shows a straight line L drawn on a coordinate grid.



(a) Find the gradient of line L.

..... **[2]**

(b) Write down the coordinates of the y-intercept of line L.

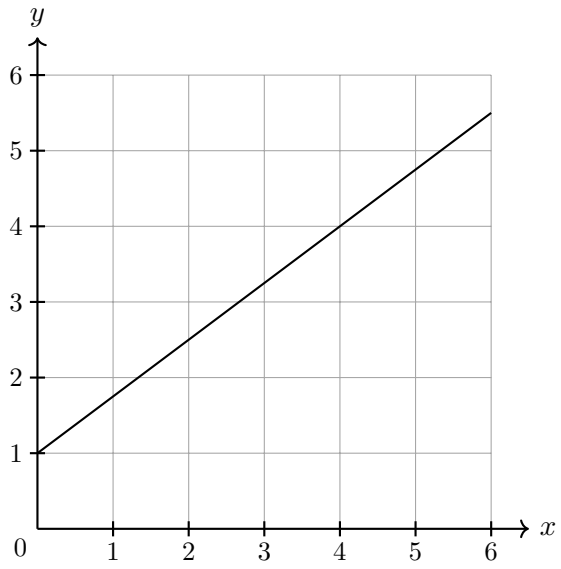
(..... , ...) **[1]**

3 The points A and B have coordinates $(-2, 8)$ and $(4, -4)$ respectively.

(a) Work out the gradient of the line AB.

..... **[2]**

(b) A straight line L is drawn on the grid.
Find the gradient of line L.



..... **[2]**

4 **(a)** Point P has coordinates $(2a, 3a)$ and point Q has coordinates $(5a, a + 4)$.

Find an expression for the gradient of the line PQ in terms of a.
Give your answer in its simplest form.

..... **[2]**

(b) The line PQ is perpendicular to a line with gradient 3.
Work out the value of a.

$a =$ ..... **[3]**

(c) The points $A(-1, 2)$, $B(3, 8)$ and $C(c, 23)$ lie on a straight line.
Work out the value of c.

$c =$ ..... **[3]**

5 The line L_1 passes through the points $P(-2, 3a)$ and $Q(4, a + 6)$.

(a) Find an expression for the gradient of L_1 in terms of a.
Give your answer in its simplest form.

..... **[3]**

(b) The line L_2 has equation $y = 3x - 5$.
Given that L_1 is parallel to L_2, find the value of a.

$a =$ ..... **[2]**

(c) Using your value of a, find the coordinates of the midpoint of PQ.

(..... , ...) **[3]**

6 The points A, B and C have coordinates $(1, 3)$, $(4, 9)$ and $(8, y)$ respectively.

(a) The points A, B and C lie on the same straight line.
Work out the value of y.

$y =$ ..... **[2]**

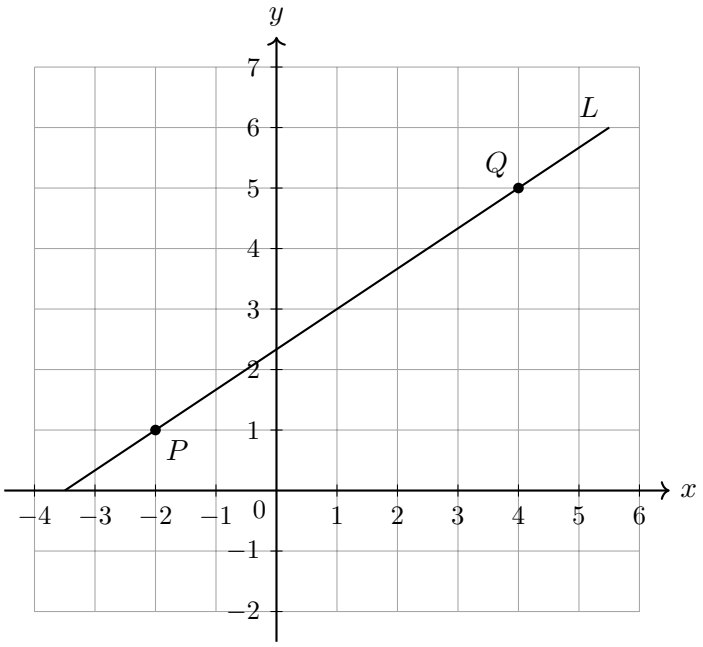
(b) A different point D has coordinates $(p, p^2 - p - 3)$.
The gradient of the line AD is 2.
Work out the possible values of p.

$p =$ ..... , $p =$ ..... **[4]**

(c) A line L has gradient m.
The line L passes through the points $(3, m)$ and $(m + 1, 12)$.
Find the two possible values of m.

$m =$ ..... , $m =$ ..... **[4]**

7 **(a)** The diagram shows a straight line L passing through the points $P(-2, 1)$ and $Q(4, 5)$.
Find the gradient of line L.



..... **[2]**

(b) A different line passes through the points $(3, k)$ and $(8, -2)$.
The gradient of this line is -1.4.
Calculate the value of k.

$k =$ ..... **[3]**

8 Point A has coordinates $(-3, 5)$ and point B has coordinates $(4, -9)$.

(a) Calculate the gradient of the line AB.

..... **[2]**

(b) Point C has coordinates $(k, 1)$.
The gradient of the line AC is equal to the gradient of the line AB.
Find the value of k.

$k =$ ..... **[3]**

9 Point P has coordinates $(k-3, 2k)$ and point Q has coordinates $(2, 5)$.

(a) The gradient of the line PQ is -3.
Work out the value of k.

$k =$ ..... **[3]**

(b) Using your value of k from part (a), find the coordinates of the midpoint of PQ.

(..... , ...) **[2]**